104. With reference to the work of human kidney, consider the following statements:

1. **After the waste is removed in the kidney, the cleaner blood is sent back through renal artery.**
2. **From Bowman's capsule, the filtered liquid passes through tiny tubes where much of the glucose is reabsorbed and sent back to the blood in the renal vein.**

Which of these statements is/are correct?

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (b) : A nephron consists of a network of tiny blood vessels, the glomerulus surrounded by Bowman's capsule. Pressure forces much of the blood plasma through the glomerulus and into Bowman's capsule. The resulting tubular fluid, which contains water and dissolved chemicals then passes into convoluted tubule and loop of Henle. This capillary network provides a mean for pickup and delivery of materials that are reabsorbed and secreted by the kidney.

105. Which one of the following statements is not correct?

(a) Milk contains none of the B-vitamins.
(b) Vitamin-A (retinol) deficiency leads to dry and scaly skin.
(c) One of the symptoms of scurvy is pain in the joints.
(d) Vitamin B_1 (thiamine) deficiency can lead to heart failure.

Ans. (a) : Due to absence of vitamin A (retinol), the skin becomes dry and night blindness occurs. Vitamins found in milk are - vitamin A, B2, B3, B7, B12, D, E.

106. With reference to normal human beings, consider the following statements:

1. **In response to the presence of HCl, secretin is produced from the duodenum.**
2. **Enterogastrone is produced in the small intestine in response to the presence of fatty acids.**

Which of these statements is/are correct?

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : When the food reaches the stomach HCl is recreated, it provide an acidic medium for pepsin to work, and kills most of the bacterial present in the food. When the food enters the duodenum (small intestine), recretin is produced by intestine juice.
Also Enterogastrone is produced in small intestine.

ENVIRONMENT & ECOLOGY

107. Consider the following statements:

1. **Molasses is a by-product of sugar production process.**
2. **Bagasse obtained in the sugar mills is used as a fuel in the boilers to generate steam for the sugar factories.**
3. **Sugar can only be produced from sugarcane as the raw material.**

Which of these statements are correct?

(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

Ans. (a) : To make molasses, sugar cane is harvested and stripped of leaves. Its juice is extracted, usually by cutting, crushing, or mashing. The juice is boiled to concentrate it, promoting sugar crystallization. For every 100 tons of Sugarcane crushed, a Sugar factory produces nearly 30 tons of wet Bagasse. Bagasse is often used as a primary fuel source for Sugar mills; when burned in quantity, it produces sufficient heat and electrical energy to supply all the needs of a typical Sugar mill, with energy to spare. Sugar could be produced from sugarcane and sugar beet, Out of 120 nations, nearly 65 nations produce from sugarcane, nearly 40 are from sugar beet and 10 are from both.

108. Which one among the following covers the highest percentage of forest area in the world?

(a) Temperate Coniferous forests
(b) Temperate Deciduous forests
(c) Tropical Monsoon forests
(d) Tropical Rain forests

Ans. (a) : Temperate Coniferous forests cover the highest percentage of forest cover in the world. Temperate coniferous forest is a terrestrial biome found in temperate regions of the world with warm summers and cool winters and adequate rainfall to sustain a forest.

109. Consider the following animals of India:
1. Crocodile 2. Elephant
Which of these is/are endangered species?

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : Crocodile and Elephant both are endangered species.

110. Which one of the following is produced during the formation of photochemical smog?

(a) Hydrocarbons
(b) Nitrogen Oxides
(c) Ozone
(d) Methane

Ans. (b) : photochemical smog formed by a complex series of chemical reactions involving sunlight oxide of nitrogen and volatile organic compounds that are present in atmosphere as a result for air pollution.
Note : During the photochemical smog no and O_3 both but major production NO_2 (Nitrogen Oxides).

CURRENT

111. With reference to India, which one of the following statements is NOT correct?

(a) IPCL is India's largest petrochemical company.
(b) RIL is the largest private sector company in India.
(c) MTNL is listed on NYSE.
(d) BSNL is the first telecom service organization in India to launch as nationwide cellular service at one time.

Ans. (a) : Jamnagar in Gujarat is the largest petrochemical industrial center of India. The city is famous for the petrochemical industry due to the presence of Reliance Ltd which is the biggest petrochemical industry in India at present. RIL is India's largest private sector company with a leadership position in the petrochemicals industry while IPCL (Indian Petrochemical Corporation Limited) is India's second largest company in that sector.

112. The Kelkar proposals which were in the news recently were the:

(a) Recommendations for reform in the power sector.
(b) Recommendations for tax reforms.
(c) Guidelines for the privatization of public sector undertakings.
(d) Guidelines for reducing vehicular pollution, and the promotion of CNG use.

Ans. (b) : The Vijay Kelkar Committee on Direct Taxes first submitted its report in year 2003.

113. Dilip Kaur Tiwana is well known as a

(a) Literateur (b) Musician
(c) Scientist (d) Sportsperson

Ans. (a) : Dr. Dalip Kaur Tiwana is a foremost novelist and short-story writer of contemporary Punjabi literature.

114. Match List I with List II and select the correct answer using the codes given below the lists:

	List I (Cricketers)		List II (Country)
A.	Barry Richards	1.	England
B.	Tex Dexter	2.	West Indies
C.	Alan Davidson	3.	South Africa
D.	Charlie Griffith	4.	New Zealand
		5.	Australia

	A	B	C	D
(a)	3	2	5	1
(b)	5	1	4	2
(c)	3	1	5	2
(d)	5	2	4	1

Ans. (c) : Barry Richards was a famous batsmen of South Africa, Tex Dexter was hard hitting batsmen of England, Alan Davidson was a renowned wicket keeper of Australia and Charlie Griffth was a fast bowler of West Indies.

115. Which one among the following has the largest shipyard in India?

(a) Kolkata (b) Mumbai
(c) Kochi (d) Visakhapatnam

Ans. (c) : Cochin Shipyard Limited (Kochi) (CSL) is the largest shipbuilding and maintenance facility in India. It is part of a line of maritime-related facilities in the port-city of Kochi, Kerala. Services provided by the shipyard are building platform supply vessels and double-hulled oil tankers.

116. Match List I with List II and select the correct answer using the codes given below the lists:

	List-I (Organisation)		List-II (Location)
A.	National Sugar Institute	1.	Dehradun
B.	Mishra Dhatu Nigam Ltd.	2.	Kamptee
C.	Institute of Military Law	3.	Pune
D.	Institute of National Integration	4.	Hyderabad
		5.	Kanpur

	A	B	C	D
(a)	3	1	4	2
(b)	5	1	4	3
(c)	3	4	5	2
(d)	5	4	2	3

Ans. (d) : Correct match of List I and List II is –

List I	List II
National Sugar Institute -	Kanpur
Mishra Dhatu Nigam Ltd. -	Hyderabad
Institute of Military Law -	Kamptee
Institute of National Integration -	Pune

117. Tim Montgomery set a new world record at the IAAF Grand Prix final in the year 2002 in.

(a) Long Jump (b) 110 m Hurdle Race
(c) 100 m Dash (d) High Jump

Ans. (c) : Tim Montgomery set a new world record at the IAAF Grand Prix Final in the year 2002 in 100 m Dash.

118. Consider the following statements:

1. **India launched its first full-fledged meteorological satellite (METSAT) in September, 2002.**
2. **For the first time, the space vehicle PSLV-C4 carried a payload of more than 1000 kg into a geosynchronous orbit.**